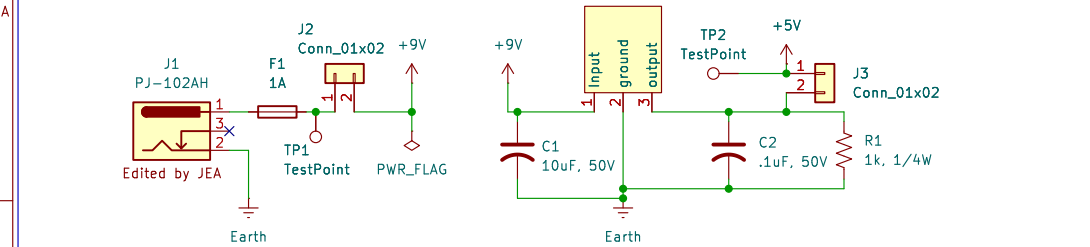


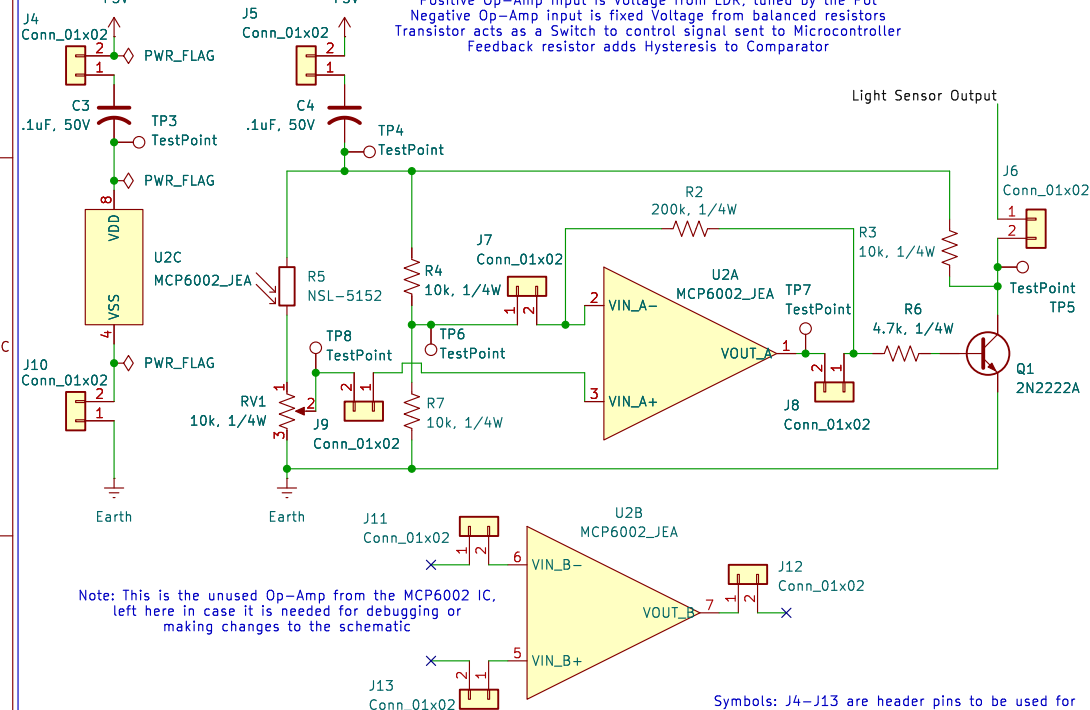
The left diagram shows the PWR_FLAG pin connected to a +9V supply through a 1A fuse (F1) and a test point (TP1). The right diagram shows the PWR_FLAG pin connected to a +5V supply through a 1k resistor (R1) and a test point (TP2). Both diagrams include a 10uF capacitor (C1) and a .1uF capacitor (C2) connected to ground.



Symbols: J2, J3 are header pins to be used for debugging, jumping, and seperating sections of the schematic in case errors occur.

Wheatstone Bridge Light Sensor

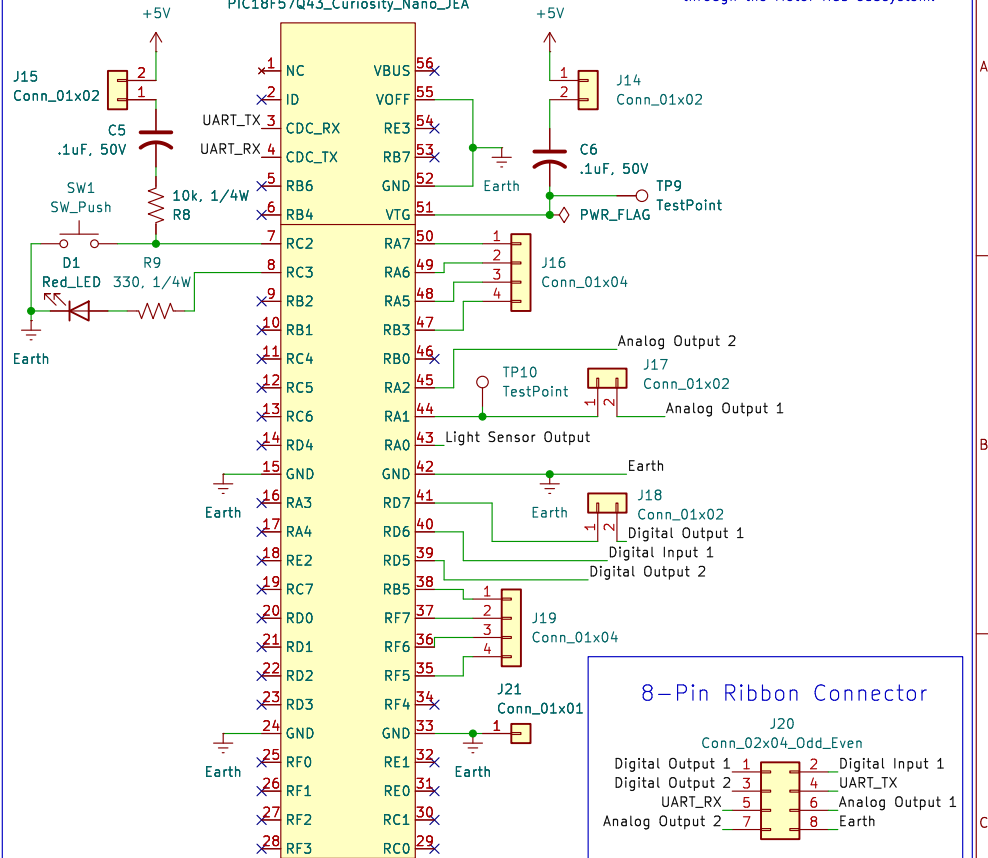
Negative Op-Amp input is fixed Voltage from balanced resistors
Transistor acts as a Switch to control signal sent to Microcontroller
Feedback resistor adds Hysteresis to Comparator



Note: This is the unused Op-Amp from the MCP6002 IC, left here in case it is needed for debugging or making changes to the schematic

Conn_01x02  Symbols: J4-J13 are header pins to be used for debugging, jumping, and separating sections of the schematic in case errors occur.

	4	5	6
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[illegible]

1

8-Pin Ribbon Connector

J20
Conn_02x04_Odd_Even

Digital Output 1	1	2	Digital Input 1
Digital Output 2	3	4	UART_TX
UART_RX	5	6	Analog Output 1
Analog Output 2	7	8	Earth

Note: Symbol J20 is the 8-Pin Connector which will connect to the other subsystems as designated by the block diagram in the datasheet

J20
Conn_02x04_Odd_Even

Digital Output 1	1	2	Digital Input 1
Digital Output 2	3	4	UART_TX
UART_RX	5	6	Analog Output 1
Analog Output 2	7	8	Earth

Note: Symbol J20 is the 8-Pin Ribbon Connector which will connect to the other subsystems as designated by the block diagram in the datasheet

Symbols: J14–J19, J21 are header pins to be used for debugging, jumping, and separating sections of the schematic in case errors occur.

Team 201: Light Sensor Subsystem

Sheet: /

File: subsystem_design.kicad_sch

Title: Wheatstone Bridge Light Sensor with Microcontroller Subsystem

Size: A4	Date: 2025-10-31	Rev: 4
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Rev: 4

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